

A hexagonal-lattice proof of the triangular-array conjecture A195806

Exact enumeration, a minimal recurrence, and a two-bound refinement

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Abstract

We count triangular arrays of side length five with entries in $\{0, \dots, n\}$, zero corners, and equal sums on lines of equal length in the three lattice directions. The explicit quasipolynomial and constant-coefficient recurrence for this count are labeled conjectural in OEIS A195806 and originate in work of Kauers and Koutschan. We give an elementary proof. An integral parametrization reduces the twelve entries to two difference parameters and four independent translations. Their admissible ranges are measured by two hexagonal norms. A common dihedral symmetry reduces the count to a weighted sum over a single chamber, with weights 1, 6, and 12. Summing this formula gives a reduced rational generating function and proves all six conjectured residue-class formulas. The minimal quasiperiod is six, the minimal eventual constant-coefficient recurrence has order fourteen, and the leading coefficient is $65/648$. We also derive a bivariate generating function for separate bounds on two classes of entries and compute its piecewise-polynomial leading volume. The report distinguishes the correctly indexed OEIS formulas from an indexing shift and a numerical typo in the printed conjecture. Reproducible exact polynomial certificates and independent finite enumerations accompany the proof.

Research status. This manuscript supplies a complete mathematical argument for the explicit OEIS formulas, rather than a conjecture inferred from additional terms. The source search described below did not establish first-in-literature priority. The manuscript has not been externally refereed or verified in a proof assistant. Its computer checks are supplementary to the proof.

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1 The problem and the precise claim being proved

1.1 Definition and indexing

Label the twelve noncorner entries of a triangular array as follows:

$$\begin{array}{ccccccc}
 & & & 0 & & & \\
 & & a & & b & & \\
 & c & & d & & e & \\
 f & & g & & h & & i \\
 0 & j & & k & & l & 0
 \end{array} . \tag{1.1}$$

The three directions are the horizontal rows and the two families of sloping lines. For each fixed line length, the three line sums must agree. Lines of *different* lengths are not required to have the same sum. In the notation of (1.1), the nontrivial equations are exactly

$$a + b = f + j = i + l, \tag{1.2}$$

$$c + d + e = c + g + k = e + h + l, \tag{1.3}$$

$$f + g + h + i = b + d + g + j = a + d + h + l, \tag{1.4}$$

$$a + c + f = j + k + l = b + e + i. \tag{1.5}$$

The three lines of length one are the prescribed zero corners, so their condition is automatic.

Definition 1.1. For an integer $n \geq 0$, let $A(n)$ be the number of arrays (1.1) satisfying (1.2)–(1.5), with all twelve letters in $\{0, \dots, n\}$.

Thus $A(0) = 1$. The OEIS entry has offset one and begins with $A(1) = 16$; we adjoin the natural value at zero to simplify generating functions. The first few counts are

$$A(0), A(1), \dots, A(7) = 1, 16, 105, 496, 1759, 5052, 12469, 27412.$$

This distinction between the entry bound and the position in a displayed sequence will matter when comparing formulas.

1.2 Why this is a documented conjecture

R. H. Hardin introduced the sequence in 2011. As retrieved on September 20, 2026, its OEIS entry explicitly labels both a recurrence and a six-part quasipolynomial as conjectured, attributing them to Manuel Kauers and Christoph Koutschan [1]. Their 2023 paper includes the corresponding problem in Section 5.1 and an explicit formula as Conjecture 11 [2, pp. 24–25].

The 2023 paper already establishes that the sequence is a quasipolynomial and hence is D-finite. The unresolved point identified there is the explicit formula, not the bare existence of some recurrence. Accordingly, the result of this report is a proof of the particular OEIS quasipolynomial and its constant-coefficient recurrence, together with structural refinements. Targeted searches for the sequence identifier and proof-related terms did not locate a later resolution; this is evidence about the sources inspected, not an exhaustive priority claim.

There is a small but important source issue. The displayed expression in Conjecture 11 uses a one-step shift relative to our entry-bound indexing, and one of its constants has a numerical typo. Section 12 gives an exact audit. Our principal target is the correctly indexed collection of formulas on the OEIS page, not the erroneous literal printed expression.

1.3 The route to a proof

A direct enumeration examines $(n+1)^{12}$ fillings. General lattice-point methods can in principle compute the answer, but the special linear system has a much smaller description. We first solve it over the integers, then count four independent intervals of translations. The remaining two-dimensional sum has a symmetry common to both interval lengths. Restricting to a symmetry chamber turns all maxima into linear functions. The generating function then requires only geometric series and two differentiations.

This order of operations is essential. Interpolating six degree-six polynomials from numerical values would not, by itself, prove that they count the arrays. Here the all-index counting formula is established before any coefficient extraction.

2 Main enumeration theorem

Theorem 2.1. *The ordinary generating function $F(z) = \sum_{n \geq 0} A(n)z^n$ is*

$$F(z) = \frac{P(z)}{(1-z)^7(1+z)(1+z+z^2)^3} = \frac{P(z)}{(1-z)^3(1-z^2)(1-z^3)^3}, \quad (2.1)$$

where

$$\begin{aligned} P(z) = & 1 + 13z + 59z^2 + 212z^3 + 471z^4 + 753z^5 + 882z^6 \\ & + 753z^7 + 471z^8 + 212z^9 + 59z^{10} + 13z^{11} + z^{12}. \end{aligned} \quad (2.2)$$

For $n = 6k + r$, where $k \geq 0$ and $0 \leq r \leq 5$, write $A(6k + r) = \sum_{d=0}^6 c_{r,d}k^d$. The coefficients are those in Table 1. In particular, the conjectured OEIS quasipolynomial is correct for every index in its stated domain.

r	$c_{r,6}$	$c_{r,5}$	$c_{r,4}$	$c_{r,3}$	$c_{r,2}$	$c_{r,1}$	$c_{r,0}$
0	4680	4680	2275	650	158	25	1
1	4680	9360	8125	3900	1133	198	16
2	4680	14040	17875	12350	4922	1087	105
3	4680	18720	31525	28600	14783	4148	496
4	4680	23400	49075	55250	35258	12121	1759
5	4680	28080	70525	94900	72197	29474	5052

Table 1: Exact coefficients of $A(6k + r)$, with powers in descending order. These agree with the conjectured OEIS formulas [1]; the final row is written in expanded form.

For example, the final row can equivalently be written in the factored form

$$A(6k + 5) = (k + 1)^2(4680k^4 + 18720k^3 + 28405k^2 + 19370k + 5052). \quad (2.3)$$

The proof of Theorem 2.1 occupies Sections 3 through 7. The period and recurrence are treated afterward, so that their minimality does not have to be assumed in deriving the formula.

3 An integral six-parameter description

Lemma 3.1. *All integer solutions of (1.2)–(1.5), without entry bounds, are given uniquely by six integers $(x, y, \alpha, \beta, \gamma, \delta)$ through*

$$\begin{array}{ccccccc}
 & & & 0 & & & \\
 & & & \beta + y & & \alpha + x - y & \\
 & \delta + 2x - y & & \gamma & & \delta + x + y & \\
 \alpha & & & \gamma + x + y & & \gamma + 2x - y & \beta \\
 0 & \beta + x & & \delta & & \alpha + x & 0
 \end{array} . \quad (3.1)$$

The inverse map is

$$x = j - i, \quad y = a - i, \quad \alpha = f, \quad \beta = i, \quad \gamma = d, \quad \delta = k. \quad (3.2)$$

Proof. Set $\alpha = f$, $\beta = i$, $\gamma = d$, and $\delta = k$. From $f + j = i + l$ we obtain

$$j - i = l - f =: x,$$

so $j = \beta + x$ and $l = \alpha + x$. Set $y = a - i$; then $a = \beta + y$. The remaining length-two equality determines

$$b = f + j - a = \alpha + x - y.$$

Next, the common length-five sum is

$$j + k + l = \alpha + \beta + \delta + 2x.$$

Equating this with $a + c + f$ and $b + e + i$ gives, respectively,

$$c = \delta + 2x - y, \quad e = \delta + x + y.$$

The length-three equations now yield

$$g = d + e - k = \gamma + x + y, \quad h = c + d - k = \gamma + 2x - y.$$

Every entry has therefore been forced into the form (3.1). This proves necessity and uniqueness.

Conversely, substitute arbitrary integers into (3.1). The three sums within each length class become, respectively,

line length	common sum
2	$\alpha + \beta + x$
3	$2\delta + \gamma + 3x$
4	$\alpha + \beta + 2\gamma + 3x$
5	$\alpha + \beta + \delta + 2x.$

All required equalities hold, including the length-four equations not used in the elimination. The inverse formulas are integral, so no congruence conditions have been lost. $\square$

Remark 3.2. The unconstrained real solution space has dimension six, and its integer lattice is identified with $\mathbb{Z}^6$ by the same parametrization. The dimension statement alone would not suffice for counting: a merely rational parametrization could hide divisibility restrictions. The explicit integral inverse in (3.2) is the point that rules them out.

4 Translation intervals and two hexagonal norms

For a real number t , write $t_+ = \max(t, 0)$.

Lemma 4.1. *For integers u, v and $n \geq 0$, the number of integers t for which $t, t + u, t + v$ all belong to $\{0, \dots, n\}$ is*

$$(n + 1 - \max(|u|, |v|, |u - v|))_+. \quad (4.1)$$

Proof. The allowed interval for t is

$$-\min(0, u, v) \leq t \leq n - \max(0, u, v).$$

Its number of integer points is the positive part of its upper endpoint minus its lower endpoint plus one. The difference between the largest and smallest of three real numbers is the largest of their three pairwise distances. Applied to $0, u, v$, this gives (4.1). $\square$

Define

$$H_1(x, y) = \max\{|x|, |y|, |x - y|\}, \quad (4.2)$$

$$H_2(x, y) = \max\{|x + y|, |2x - y|, |x - 2y|\}. \quad (4.3)$$

Both are norms on $\mathbb{R}^2$ with centrally symmetric hexagonal unit balls. Only their explicit maximum formulas are needed below.

For fixed x, y , the four translation parameters in (3.1) occur in four disjoint triples. Their offsets are

translation	offsets	range
β	$0, y, x$	$H_1(x, y)$
α	$0, x - y, x$	$H_1(x, y)$
γ	$0, x + y, 2x - y$	$H_2(x, y)$
δ	$0, 2x - y, x + y$	$H_2(x, y)$.

(4.4)

The first two ranges agree because their pairwise differences, up to sign, are $x, y, x - y$. The last two triples simply have their nonzero offsets interchanged.

Proposition 4.2. *For every $n \geq 0$,*

$$A(n) = \sum_{(x,y) \in \mathbb{Z}^2} (n + 1 - H_1(x, y))_+^2 (n + 1 - H_2(x, y))_+^2. \quad (4.5)$$

Proof. The parametrization is bijective, and the four translation parameters are independent once x, y have been fixed. Lemma 4.1 therefore gives the product in each summand. A summand is nonzero only if $H_1(x, y) \leq n$, which in particular implies $|x|, |y| \leq n$. Thus the displayed sum has only finitely many nonzero terms and accounts for every bounded array exactly once. $\square$

This already gives an $O(n^2)$ -term counting formula, replacing the original twelve-dimensional search. The remaining maxima are eliminated by exploiting a symmetry shared by both norms.

5 A common dihedral chamber

Lemma 5.1. *The transformations*

$$T(x, y) = (x - y, x), \quad S(x, y) = (x, x - y) \quad (5.1)$$

generate a group of order twelve preserving $\mathbb{Z}^2$, H_1 , and H_2 . Each orbit meets the closed chamber

$$\mathcal{C} = \{(x, y) \in \mathbb{R}^2 : x \geq 2y \geq 0\} \quad (5.2)$$

in exactly one point. A nonzero chamber point has orbit size six on either boundary ray and twelve in the interior; the origin has orbit size one.

Proof. A direct computation gives $T^6 = S^2 = 1$ and $STS = T^{-1}$. For example, the three linear forms defining H_2 are transformed by T into

$$2x - y, \quad x - 2y, \quad -(x + y),$$

so their absolute values are permuted. The same check works for S and for the defining forms of H_1 .

For completeness, the fundamental-domain assertion can be seen in the positive-definite quadratic form

$$q_0(x, y) = x^2 - xy + y^2.$$

Both transformations preserve this form. In the corresponding Euclidean metric, T is a rotation through 60 degrees: its determinant is one and its trace is one. The map S is the reflection fixing the line $x = 2y$. The reflection $T^{-1}S$ fixes the line $y = 0$. The positive rays on these two lines enclose an angle of 30 degrees. Their images divide the plane into twelve sectors, whose interiors are disjoint. This proves that the group has order twelve and that the closed sector (5.2) contains exactly one representative of each orbit.

An interior nonzero point is fixed by no nonidentity group element. A point on either boundary ray is fixed by exactly one reflection, so its stabilizer has order two. The orbit sizes follow, while every group element fixes the origin. $\square$

Remark 5.2. This is a symmetry of the auxiliary parameter lattice. We are *not* counting arrays modulo rotations or reflections. All arrays remain labeled by their positions, and the orbit sizes are multiplicities in a sum, not identifications of arrays.

Use integral coordinates on the chamber by setting

$$p = x - 2y, \quad q = y, \quad \text{so that} \quad x = p + 2q, \quad y = q. \quad (5.3)$$

This is a bijection between $\mathcal{C} \cap \mathbb{Z}^2$ and $\mathbb{Z}_{\geq 0}^2$. On the chamber,

$$H_1 = p + 2q, \quad H_2 = 2p + 3q. \quad (5.4)$$

Indeed, x is the largest defining absolute value for H_1 , and $2x - y$ is the largest for H_2 . In particular,

$$\frac{3}{2}H_1 \leq H_2 \leq 2H_1 \quad (5.5)$$

holds globally, by symmetry and (5.4).

Define the multiplicity

$$\mu(p, q) = \begin{cases} 1, & p = q = 0, \\ 6, & \text{exactly one of } p, q \text{ is zero,} \\ 12, & p, q > 0. \end{cases} \quad (5.6)$$

The equal-bound count can now be written with no maxima.

Theorem 5.3. *For every integer $n \geq 0$,*

$$A(n) = \sum_{\substack{p, q, r \geq 0 \\ 2p + 3q + r = n}} \mu(p, q)(r + 1)^2(r + p + q + 1)^2. \quad (5.7)$$

Proof. Apply Lemma 5.1 to (4.5). Since $H_2 \geq H_1$, a nonzero chamber summand is characterized by $2p + 3q \leq n$. Set $r = n - 2p - 3q$. Then

$$n + 1 - H_2 = r + 1, \quad n + 1 - H_1 = r + p + q + 1.$$

The orbit multiplicities give exactly (5.7). □

The first values illustrate the boundary weights. For $n = 1$, only $p = q = 0$ occurs, so $A(1) = 2^4 = 16$. For $n = 2$, one new point occurs on the $q = 0$ boundary:

$$A(2) = 3^4 + 6 \cdot 1^2 \cdot 2^2 = 105.$$

For $n = 3$, the two nonzero possibilities are $(p, q, r) = (1, 0, 1)$ and $(0, 1, 0)$, giving

$$A(3) = 4^4 + 6 \cdot 2^2 \cdot 3^2 + 6 \cdot 1^2 \cdot 2^2 = 496.$$

These examples are checks on the formula; its proof has no finite-index restriction.

6 Summing the chamber formula

Let

$$G(u, v) = \sum_{p, q \geq 0} \mu(p, q) u^p v^q. \quad (6.1)$$

Separate the origin, the two axes, and the open quadrant to obtain

$$\begin{aligned} G(u, v) &= 1 + \frac{6u}{1-u} + \frac{6v}{1-v} + \frac{12uv}{(1-u)(1-v)} \\ &= \frac{1 + 5u + 5v + uv}{(1-u)(1-v)}. \end{aligned} \quad (6.2)$$

The Euler operator

$$D = u \frac{\partial}{\partial u} + v \frac{\partial}{\partial v}$$

multiplies the coefficient of $u^p v^q$ by $p + q$. Hence DG and D^2G encode the first two powers of that sum. Direct differentiation gives

$$DG = \frac{6(u+v)(1-uv)}{(1-u)^2(1-v)^2}, \quad (6.3)$$

$$D^2G = \frac{6(u+v)(1+u+v-6uv+u^2v+uv^2+u^2v^2)}{(1-u)^3(1-v)^3}. \quad (6.4)$$

For $j \geq 0$, put $S_j(z) = \sum_{r \geq 0} (r+1)^j z^r$. The needed cases are

$$\begin{aligned} S_2(z) &= \frac{1+z}{(1-z)^3}, & S_3(z) &= \frac{1+4z+z^2}{(1-z)^4}, \\ S_4(z) &= \frac{1+11z+11z^2+z^3}{(1-z)^5}. \end{aligned} \quad (6.5)$$

These follow from the geometric series by applying $z d/dz + 1$ repeatedly. The weight in (5.7) expands as

$$(r+1)^2(r+p+q+1)^2 = (r+1)^4 + 2(p+q)(r+1)^3 + (p+q)^2(r+1)^2.$$

Multiplying by $z^{2p+3q+r}$ and summing therefore yields the exact formal power-series identity

$$F(z) = (S_4(z)G(u, v) + 2S_3(z)DG(u, v) + S_2(z)D^2G(u, v)) \Big|_{u=z^2, v=z^3}. \quad (6.6)$$

Each coefficient involves finitely many triples, so there is no analytic interchange-of-limits issue.

6.1 An explicit certificate for the rational simplification

To make the calculation from (6.6) to (2.1) checkable without trusting a computer algebra simplifier, set $U = z^2$, $V = z^3$, and define

$$\begin{aligned} M_0 &= 1 + 5U + 5V + UV, \\ M_1 &= 6(U+V)(1-UV), \\ M_2 &= 6(U+V)(1+U+V-6UV+U^2V+UV^2+U^2V^2). \end{aligned}$$

The common numerator on the right side of (6.6), over the denominator $(1-z)^5(1-U)^3(1-V)^3$, is

$$\begin{aligned} &(1+11z+11z^2+z^3)M_0(1-U)^2(1-V)^2 \\ &+ 2(1+4z+z^2)(1-z)M_1(1-U)(1-V) \\ &+ (1+z)(1-z)^2M_2 = (1-z)^2(1-U)^2P(z). \end{aligned} \quad (6.7)$$

Equation (6.7) is an identity between integer polynomials of degree at most eighteen. It is verified simply by expanding and comparing coefficients; the supplied standard-library verifier performs exactly that multiplication and comparison.

Canceling the displayed factors gives

$$F(z) = \frac{P(z)}{(1-z)^3(1-U)(1-V)^3}.$$

Substitute $U = z^2$, $V = z^3$ and factor $1 - z^2$ and $1 - z^3$ to obtain (2.1). This establishes the generating-function part of Theorem 2.1 directly from the combinatorial definition.

7 Exact coefficient extraction

7.1 Polynomial and periodic parts

A compact form of the answer separates a polynomial in the centered variable $N = n + 1$ from periodic corrections. Define

$$M(N) = \frac{65N^6}{648} + \frac{325N^4}{1296} + \frac{439N^2}{324} - \frac{2521}{7776} \quad (7.1)$$

and

$$\begin{aligned} B_0(N) &= \frac{2(3N^2 - 24N - 14)}{243}, \\ B_1(N) &= \frac{2(3N^2 + 24N - 14)}{243}, \\ B_2(N) &= -\frac{4(3N^2 - 14)}{243}. \end{aligned} \quad (7.2)$$

Proposition 7.1. *For every $n \geq 0$,*

$$\boxed{A(n) = M(n+1) - \frac{3}{32}(-1)^n + B_{n \bmod 3}(n+1).} \quad (7.3)$$

Proof. Write $C = 1 + z + z^2$. Partial fractions of (2.1) give

$$\begin{aligned} F(z) &= \frac{650}{9(1-z)^7} - \frac{1625}{9(1-z)^6} + \frac{325}{2(1-z)^5} - \frac{2275}{36(1-z)^4} \\ &\quad + \frac{2687}{216(1-z)^3} - \frac{737}{432(1-z)^2} - \frac{2521}{7776(1-z)} - \frac{3}{32(1+z)} \\ &\quad - \frac{14(4z-1)}{243C} - \frac{2(13z+8)}{81C^2} + \frac{4(z-1)}{27C^3}. \end{aligned} \quad (7.4)$$

This identity can be checked by multiplying by $7776(1-z)^7(1+z)C^3$; the resulting integer-polynomial equality is a second directly checkable algebra calculation, included in the verifier.

The identity

$$[z^n](1-z)^{-j} = \binom{n+j-1}{j-1}$$

shows that the first seven terms in (7.4) have coefficient $M(n+1)$. The $(1+z)^{-1}$ term contributes $-3(-1)^n/32$.

For the last three terms, use

$$C^{-j} = \frac{(1-z)^j}{(1-z^3)^j}, \quad (1-z^3)^{-j} = \sum_{k \geq 0} \binom{k+j-1}{j-1} z^{3k}. \quad (7.5)$$

Expanding the short numerator $(1-z)^j$ and separating the three residue classes gives the following coefficients for their sum:

n	coefficient of the last three terms of (7.4)
$3k$	$(54k^2 - 108k - 70)/243$
$3k+1$	$(54k^2 + 216k + 92)/243$
$3k+2$	$(-108k^2 - 216k - 52)/243$.

These are exactly $B_{n \bmod 3}(n+1)$ in (7.2). All extractions hold also at $k=0$, either directly from (7.5) or by substituting into the three displayed polynomials. This proves (7.3). $\square$

Substitute $n = 6k + r$ into (7.3). The parity and residue modulo three are then fixed, so ordinary polynomial expansion gives each row of Table 1. This completes the proof of Theorem 2.1. Notice that the periodic corrections have degree at most two. Consequently, the coefficients of n^6, n^5, n^4, n^3 do not depend on the residue class.

7.2 An independent all-index certificate for the residue table

The table can also be verified directly against the rational function, without using the partial fraction decomposition. Let polynomials $T_d(t)$ be defined by

$$T_0(t) = 1, \quad T_{d+1}(t) = t((1-t)T'_d(t) + (d+1)T_d(t)). \quad (7.6)$$

Differentiating a geometric series shows that

$$\sum_{k \geq 0} k^d t^k = \frac{T_d(t)}{(1-t)^{d+1}}$$

for $d \geq 1$; for $d=0$ the left side is understood as $\sum_{k \geq 0} t^k$. Define the integer polynomial

$$H(z) = \sum_{r=0}^5 z^r \sum_{d=0}^6 c_{r,d} T_d(z^6) (1-z^6)^{6-d}. \quad (7.7)$$

The generating function of the residue table is $H(z)/(1-z^6)^7$. Let

$$Q(z) = (1-z)^7(1+z)(1+z+z^2)^3. \quad (7.8)$$

A third finite integer-polynomial identity is

$$H(z)Q(z) = P(z)(1-z^6)^7. \quad (7.9)$$

The degree on both sides is at most fifty-four. The included program constructs T_d , H , P , and Q as coefficient lists and checks this identity exactly. It certifies equality of the *whole* generating functions, not merely agreement of a finite prefix. Since their denominators have constant term one, (7.9) proves coefficientwise equality in $\mathbb{Z}[[z]]$.

8 Minimal period, minimal recurrence, and growth

Proposition 8.1. *The fraction in (2.1) is reduced over $\mathbb{Q}[z]$.*

Proof. Direct evaluation and polynomial division give

$$P(1) = 3900, \quad P(-1) = -12, \quad P(z) \equiv 12 \pmod{1+z+z^2}. \quad (8.1)$$

These are all irreducible factors of the denominator, and none divides P . $\square$

Corollary 8.2. *The sequence $A(n)$ has degree six and minimal quasiperiod six.*

Proof. Table 1 supplies quasiperiod six and degree six. Suppose a quasipolynomial representation had quasiperiod h . Summing each of its polynomial residue classes would express its generating function with poles only at h th roots of unity: a common denominator is a power of $1 - z^h$. But the reduced generating function has a pole at -1 and at both primitive cube roots of unity by (8.1). Thus $2 \mid h$ and $3 \mid h$, so $6 \mid h$. No smaller positive quasiperiod is possible. $\square$

Let E denote the forward shift on sequences, $(EA)(n) = A(n+1)$.

Corollary 8.3. *For every $n \geq 0$,*

$$(E-1)^7(E+1)(E^2+E+1)^3A(n) = 0. \quad (8.2)$$

The minimal order of an eventual homogeneous linear recurrence with constant coefficients over $\mathbb{Q}$ is fourteen.

Proof. The generating function has denominator degree fourteen and numerator degree twelve. Multiplication by $Q(z)$ gives $Q(z)F(z) = P(z)$, so every coefficient of degree at least fourteen vanishes. Reversing the denominator gives the forward-shift recurrence (8.2). In fact $z^{14}Q(1/z) = -Q(z)$, so the sign change is immaterial.

Conversely, any eventual homogeneous constant-coefficient recurrence of order d yields a polynomial multiplier of degree at most d that makes $F(z)$ a polynomial: the recurrence cancels the entire sufficiently long tail. Since P/Q is reduced, that multiplier must be divisible by Q . Therefore $d \geq \deg Q = 14$. $\square$

The qualifier *constant coefficients* is important. This result is not a minimality claim among recurrences with coefficients depending polynomially on n . The 2023 source reports a smaller-order polynomial-coefficient guess; it is a different notion of order [2]. The constant-coefficient recurrence conjectured on the OEIS page is precisely the expanded form

$$\begin{aligned} 0 = & A(n) - 3A(n+1) + 2A(n+2) - A(n+3) + 6A(n+4) \\ & - 5A(n+5) - 3A(n+6) + 3A(n+8) + 5A(n+9) \\ & - 6A(n+10) + A(n+11) - 2A(n+12) + 3A(n+13) - A(n+14). \end{aligned} \quad (8.3)$$

There is no $A(n+7)$ term.

Corollary 8.4. *As $n \rightarrow \infty$,*

$$A(n) = \frac{65}{648}n^6 + \frac{65}{108}n^5 + \frac{2275}{1296}n^4 + \frac{325}{108}n^3 + O(n^2). \quad (8.4)$$

In particular, $A(n) \sim 65n^6/648$.

Proof. Expand $M(n+1)$ in (7.3); the remaining terms have size $O(n^2)$. Independently, the leading constant follows from the pole at one:

$$\lim_{z \rightarrow 1} (1-z)^7 F(z) = \frac{P(1)}{2 \cdot 3^3} = \frac{650}{9},$$

and the coefficient of $(1-z)^{-7}$ has leading term $n^6/6!$. $\square$

The centered exact formula is more informative than an asymptotic expansion: it records every lower-order oscillation, including the constant alternating term. No exponentially small or uncomputed remainder is needed for exact evaluation.

9 Reciprocity and interior arrays

The numerator has the palindromic symmetry $z^{12}P(1/z) = P(z)$, while the denominator is antipalindromic. Consequently,

$$F(1/z) = -z^2 F(z) \quad (9.1)$$

as an identity of rational functions. This is not a termwise identification of the power-series expansions around zero and infinity.

The residue formulas uniquely extend $A(n)$ from nonnegative to all integer arguments: use the polynomial assigned to the residue class of n . Denote this continuation by $\tilde{A}(n)$.

Proposition 9.1. *For every integer n ,*

$$\tilde{A}(-n) = \tilde{A}(n - 2). \quad (9.2)$$

For $n \geq 2$, the number of valid arrays with all twelve noncorner entries strictly between 0 and n is $A(n - 2)$.

Proof. The polynomial $M(N)$ is even, and

$$B_0(-N) = B_1(N), \quad B_1(-N) = B_0(N), \quad B_2(-N) = B_2(N).$$

Under the replacement of the sequence argument t by $-t - 2$, the centered variable $N = t + 1$ changes sign, the parity of t stays fixed, and the residues $0, 1, 2$ modulo three are changed to $1, 0, 2$, respectively. Formula (7.3), regarded as a quasipolynomial identity, therefore gives $\tilde{A}(t) = \tilde{A}(-t - 2)$. Substitute $t = n - 2$.

For the counting assertion, subtract one from each noncorner entry and keep the three corners zero. Within a fixed line-length class, every line contains the same number of noncorner positions: those numbers are 2, 3, 4, 3 for lengths 2, 3, 4, 5. Hence each equality of line sums is preserved. The strict integer bounds $1 \leq a, \dots, l \leq n - 1$ become the closed bounds $0 \leq a, \dots, l \leq n - 2$. Adding one gives the inverse bijection. $\square$

The geometric interpretation is also exact. In the six coordinates of Lemma 3.1, let $\mathcal{P}$ be the real polytope obtained by bounding every noncorner entry between zero and one. The point $x = y = 0$, $\alpha = \beta = \gamma = \delta = 1/2$ satisfies every bound strictly, so $\mathcal{P}$ is six-dimensional. Its relative interior consists exactly of points where every noncorner entry is strictly between the bounds. Indeed, each coordinate-bound functional is nonconstant on the solution space, and a point attaining such a bound lies on a supporting hyperplane; conversely, finitely many strict linear inequalities remain strict in a small neighborhood.

Thus the preceding bijection describes the interior lattice-point count of $n\mathcal{P}$. At $n = 1$ there are no such points and $\tilde{A}(-1) = 0$. At $n = 2$ there is exactly one: all twelve noncorner entries are one. These statements and (9.2) have been proved directly, without needing a general reciprocity theorem as a black box.

10 Two entry bounds and a bivariate generating function

The parametrization naturally divides the entries into two classes:

$$\mathcal{U} = \{a, b, f, i, j, l\}, \quad \mathcal{V} = \{c, d, e, g, h, k\}. \quad (10.1)$$

The first class contains the six positions adjacent to corners. The second contains the three side midpoints and the three interior positions. Let $A(n, m)$ count the same arrays, with entries in $\mathcal{U}$ bounded between zero and n and entries in $\mathcal{V}$ bounded between zero and m . The corners remain zero, and $A(n, n) = A(n)$.

Theorem 10.1. *For integers $n, m \geq 0$,*

$$A(n, m) = \sum_{p, q \geq 0} \mu(p, q) (n+1-p-2q)_+^2 (m+1-2p-3q)_+^2. \quad (10.2)$$

Moreover,

$$\begin{aligned} \sum_{n, m \geq 0} A(n, m) X^n Y^m \\ = \frac{(1+X)(1+Y)}{(1-X)^3(1-Y)^3} \frac{1+5XY^2+5X^2Y^3+X^3Y^5}{(1-XY^2)(1-X^2Y^3)}. \end{aligned} \quad (10.3)$$

Proof. In (4.4), the α - and β -triples are exactly $\mathcal{U}$, while the γ - and δ -triples are exactly $\mathcal{V}$. Their independent translation counts are therefore $(n+1-H_1)_+^2 (m+1-H_2)_+^2$. The same chamber argument proves (10.2); no comparison between n and m is needed.

For a fixed chamber point, write

$$n = p + 2q + r, \quad m = 2p + 3q + s, \quad r, s \geq 0.$$

Its contribution is $\mu(p, q)(r+1)^2(s+1)^2$. Consequently the bivariate series factors as

$$S_2(X)S_2(Y)G(XY^2, X^2Y^3).$$

Substitution of (6.2) and (6.5) gives (10.3). □

As boundary checks,

$$A(n, 0) = (n+1)^2, \quad A(0, m) = (m+1)^2.$$

In either case the narrower bound forces $x = y = 0$ and leaves two independent translations for the other class.

The equal-bound generating function is the *diagonal* of the bivariate series: extract the coefficients of $X^n Y^n$ and attach z^n . It is not obtained by substituting $X = Y = z$, which would group terms by $n + m$. This distinction explains why the very short bivariate rational form can coexist with the more elaborate univariate numerator in (2.2).

11 The leading volume under unequal scaling

The two-bound formula also yields a continuous leading coefficient with an explicit change of shape. For fixed real $a, b > 0$, define

$$V(a, b) = 12 \iint_{\substack{p, q \geq 0 \\ p+2q \leq a \\ 2p+3q \leq b}} (a-p-2q)^2 (b-2p-3q)^2 dp dq. \quad (11.1)$$

Theorem 11.1. For fixed $a, b > 0$, as $t \rightarrow \infty$,

$$A(\lfloor at \rfloor, \lfloor bt \rfloor) = V(a, b)t^6 + O_{a,b}(t^5). \quad (11.2)$$

The leading coefficient has the truncated-power expression

$$V(a, b) = \frac{b^4(540a^2 - 252ab + 37b^2) - (2b - 3a)_+^6 + 27(b - 2a)_+^6}{3240}. \quad (11.3)$$

Proof of the asymptotic formula. Rescale the chamber variables in (10.2) by t . There are $O_{a,b}(t^2)$ contributing lattice points. At an interior chamber point, the multiplicity is twelve. Replacing each integer bound plus one by at or bt changes the summand by $O_{a,b}(t^3)$ on a common bounded rescaled support, hence changes the total by $O_{a,b}(t^5)$.

The two axes contain only $O_{a,b}(t)$ relevant points. Replacing their multiplicity six by twelve, and treating the origin similarly, changes the sum by $O_{a,b}(t^5)$ because each weight is $O_{a,b}(t^4)$. The remaining expression is t^6 times a two-dimensional Riemann sum for the compactly supported function

$$12(a - p - 2q)_+^2(b - 2p - 3q)_+^2 \quad \text{on the first quadrant.}$$

On a fixed rectangle containing its support this function is Lipschitz. The Riemann-sum error is $O_{a,b}(1/t)$, obtained by bounding the variation on each square of side $1/t$. Multiplying by t^6 gives the stated $O_{a,b}(t^5)$ error and the integral (11.1). $\square$

Evaluation of the integral. The two upper bounds on p are

$$p \leq a - 2q, \quad p \leq \frac{b - 3q}{2}.$$

Their intersection is

$$(p, q) = (2b - 3a, 2a - b).$$

The integration region changes combinatorial type at $b = 3a/2$ and $b = 2a$.

If $b \leq 3a/2$, only the second sloping bound is active. Integration over $0 \leq q \leq b/3$ and $0 \leq p \leq (b - 3q)/2$ yields

$$V(a, b) = \frac{b^4(540a^2 - 252ab + 37b^2)}{3240}. \quad (11.4)$$

If $b \geq 2a$, only the first sloping bound is active. Integration over $0 \leq q \leq a/2$ and $0 \leq p \leq a - 2q$ gives

$$V(a, b) = \frac{a^4(37a^2 - 84ab + 60b^2)}{120}. \quad (11.5)$$

In the intermediate region $3a/2 \leq b \leq 2a$, split at $q = 2a - b$. For $0 \leq q \leq 2a - b$, use the upper bound $(b - 3q)/2$ on p ; for $2a - b \leq q \leq a/2$, use $a - 2q$. The result is

$$V(a, b) = -\frac{1}{120}(27a^6 - 108a^5b + 180a^4b^2 - 160a^3b^3 + 60a^2b^4 - 12ab^5 + b^6). \quad (11.6)$$

These integrations involve only a polynomial of degree four, with linear endpoints, so ordinary term-by-term integration suffices. The optional symbolic companion verifies the three definite integrals exactly.

Finally, subtract (11.4) from (11.6); the difference is $-(2b - 3a)^6/3240$. Subtract (11.6) from (11.5); the difference is $(b - 2a)^6/120$. This gives (11.3) in all three regions, including their shared boundaries. $\square$

For equal bounds, the first region applies and

$$V(a, a) = \frac{65}{648}a^6,$$

agreeing with (8.4). The sixth-power corrections also show that the leading volume and its derivatives through order five match across each positive boundary ray; the change first becomes visible in sixth transverse derivatives.

There is a direct volume interpretation. For continuous entries with bounds a, b , fixed difference parameters leave four intervals of translation of lengths $a - H_1, a - H_1, b - H_2, b - H_2$, when positive. Their product is the four-dimensional fiber volume. Integrating over x, y , then using the twelve chambers and the determinant-one change (5.3), gives precisely (11.1). Hence $65/648$ is the volume of $\mathcal{P}$ in the measure for which the integer lattice in its six parametrizing coordinates has covolume one. It is *not* the convention that multiplies this volume by $6!$; under that alternative normalization the value is $650/9$.

12 Source audit: the shift and the printed constant

Let $T(n)$ denote the literal expression printed in Conjecture 11 of Kauers and Koutschan [2, p. 25]. Written schematically, it is

$$T(n) = \frac{130n^6 + 1560n^5 + 8125n^4 + 23400n^3 + u_r n^2 + v_r n + w_r}{1296}, \quad r = n \bmod 6, \quad (12.1)$$

with the following printed lower coefficients:

r	u_r	v_r	w_r
0	40788	42768	20736
1	40692	42128	20045
2	40788	42256	19712
3	40788	42768	20493
4	40692	42128	20288
5	40788	42256	19496

Both the journal PDF and the arXiv version inspected for this report display 19496 in the final row.

There are two distinct observations. First, $T(0) = 16$, while the entry-bound count is $A(0) = 1$. The beginning of the printed expression is shifted by one relative to the entry bound. Second, the final row has a genuine arithmetic inconsistency, not just a shift. At $n = 5$,

$$T(5) = \frac{598513}{48} = 12469 + \frac{1}{48}, \quad (12.2)$$

which cannot be an array count.

Our exact formulas determine the comparison for every index:

$$T(n) - A(n+1) = \begin{cases} 1/48, & n \equiv 5 \pmod{6}, \\ 0, & \text{otherwise,} \end{cases} \quad n \geq 0. \quad (12.3)$$

Indeed, substitute $n = 6k + r$ into (12.1) and compare coefficients with Table 1, using $A(6k+6) = A(6(k+1))$ for the final class. Five polynomial differences vanish; the remaining numerator difference is the constant 27, giving $27/1296 = 1/48$.

Replacing 19496 by 19469 removes that constant difference. Thus the corrected printed expression is exactly $A(n+1)$. Equivalently, substituting $n-1$ into the corrected expression gives the count with entries bounded by n , for $n \geq 1$.

This audit is not the main solution. Finding the nonintegral value in (12.2) would only refute a transcription-level statement. The substantive result is the derivation from the array constraints of the correct OEIS quasipolynomial, including every residue class and all indices.

13 Verification, reproducibility, and computational scope

The archive contains two verification programs. The main program, `verify.py`, uses only the Python standard library and exact integers or rational numbers. The optional `verify_symbolic.py` uses SymPy for independent differentiation, rational simplification, and polynomial integration. Neither program requires network access.

13.1 What was actually checked

The main program completed thirteen groups of checks. Three are all-index algebra certificates: the degree-eighteen identity (6.7), the denominator-cleared partial fractions (7.4), and the degree-fifty-four identity (7.9). It also checks the noncancellation values (8.1) and the numerator/denominator reversal identities.

For an independent combinatorial test, it constructs the three directions from triangular-grid coordinates (r, c) , $0 \leq c \leq r \leq 4$, rather than copying the parametrization into the validity test. It exhaustively examines every filling for bounds zero, one, and two. The numbers of candidate fillings are 1, 4096, and 531441, and the numbers accepted are 1, 16, and 105. Each accepted filling is also sent through the inverse parametrization and reconstructed.

The program additionally compares the full two-dimensional lattice sum with the chamber sum for $0 \leq n \leq 100$, the chamber sum with the residue formula for $0 \leq n \leq 300$, and the rational-function coefficients with the residue formula for $0 \leq n \leq 1000$. All twenty-nine initial values displayed on the retrieved OEIS page agree. It checks the auxiliary dihedral orbits on $[-20, 20]^2$, the centered formula and reciprocity for $-200 \leq n \leq 200$, and three independent descriptions of $A(n, m)$ on the rectangle $0 \leq n, m \leq 20$. The source comparison (12.3) is checked by exact polynomial composition, not only by numerical substitution.

The optional symbolic program completed nine further exact checks: the two Euler derivatives, the rational generating-function simplification, three volume integrations, two boundary-correction identities, and the equal-bound leading coefficient. The run used Python 3.13.5 and SymPy 1.14.0. Machine-readable reports record the versions and measured run times.

13.2 What these checks do and do not prove

The manuscript proves the integer bijection, translation count, fundamental domain, and summation identity for all indices. Those steps are the mathematical foundation. The polynomial certificates verify finite algebra identities that themselves imply all-index generating-function statements. The finite enumerations serve a different purpose: they independently detect mistakes in the interpretation of the array constraints, indexing, chamber weights, or copied coefficients.

A finite test is not promoted to a general proof merely because its range is large. Conversely, a finite *coefficient comparison between two explicitly bounded-degree polynomials* proves their identity. The verifier keeps these two types of evidence separate. None of the files constitutes a proof-assistant formalization, and the numerical tests cannot by themselves establish literature priority.

13.3 Rebuilding and using the artifacts

From the extracted archive directory, run

```
python3 verify.py
python3 verify_symbolic.py      # optional; requires SymPy
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The second LaTeX pass resolves the contents and cross-references. `data/sequence.csv` contains $A(n)$ for $0 \leq n \leq 1000$; `data/two_bound_counts.csv` contains the checked two-bound rectangle. `README.md` documents the files, conventions, and status. The supplied `Makefile` provides equivalent build targets.

For evaluation at a single very large nonnegative index, the residue polynomial is the most economical formula: choose the residue and use Horner's rule in $k = \lfloor n/6 \rfloor$. It uses a fixed number of integer arithmetic operations, although their bit cost grows with the size of n . The chamber sum is preferable as a transparent independent enumerator, with $O(n^2)$ terms. The order-fourteen recurrence is convenient for generating a long prefix. These are arithmetic-operation statements, not claims of constant bit complexity.

14 Conclusion and remaining directions

The explicit A195806 quasipolynomial can be derived from an elementary structure that is not apparent in the twelve-entry definition. Four translation variables account for the fourth-degree weight, while two difference variables account for the sixth-degree total growth. The two translation ranges are hexagonal norms with a common twelvefold parameter symmetry. In a fundamental chamber their values become $p + 2q$ and $2p + 3q$, which explains the appearance of the denominators $1 - z^2$ and $1 - z^3$ and, after cancellation, the exact quasiperiod six.

The main result is a proof of the correctly indexed formulas and recurrence still marked conjectural in the source inspected, together with minimality, reciprocity, and a two-bound extension. It does not resolve every sequence in the 2023 paper, establish a new general lattice-point algorithm, or prove that no earlier solution to this particular conjecture exists. Those are separate claims and are not needed here.

A natural next investigation is the analogous triangular-array problem at other side lengths. The successful part of the present method is not simply linear elimination, which remains available in general, but the subsequent discovery of repeated translation ranges and a common manageable chamber structure. Determining when those features persist is a concrete structural question. No unverified general formula for larger triangles is asserted in this report.

A Coefficient lists and a compact counting implementation

For use in independent implementations, the denominator expands as

$$Q(z) = 1 - 3z + 2z^2 - z^3 + 6z^4 - 5z^5 - 3z^6 \\ + 3z^8 + 5z^9 - 6z^{10} + z^{11} - 2z^{12} + 3z^{13} - z^{14}.$$

The numerator and denominator coefficients, in ascending order, are

P = [1,13,59,212,471,753,882,753,471,212,59,13,1]
 Q = [1,-3,2,-1,6,-5,-3,0,3,5,-6,1,-2,3,-1]

Since $Q(0) = 1$, coefficients of P/Q can be generated recursively as

$$A(n) = P_n - \sum_{j=1}^{\min(n,14)} Q_j A(n-j),$$

where $P_n = 0$ for $n > 12$. This formula incorporates the initial terms and avoids a separate recurrence initialization convention.

The core chamber enumerator is short enough to reproduce in full:

```
def count(n: int) -> int:
    if isinstance(n, bool) or not isinstance(n, int) or n < 0:
        raise ValueError("n must be a nonnegative integer")
    total = 0
    for p in range(n // 2 + 1):
        for q in range((n - 2*p) // 3 + 1):
            r = n - 2*p - 3*q
            mu = 1 if p == q == 0 else 6 if p*q == 0 else 12
            total += mu * (r+1)**2 * (r+p+q+1)**2
    return total
```

Every term is nonnegative, and no floating-point operation is involved. The complete verifier adds error checks, independent enumerators, exact polynomial arithmetic, reports, and data export.

B A short numerical reference table

n	$A(n)$	n	$A(n)$	n	$A(n)$
0	1	7	27412	14	1155567
1	16	8	55059	15	1699684
2	105	9	102952	16	2442553
3	496	10	181543	17	3438468
4	1759	11	304908	18	4752283
5	5052	12	491563	19	6460432
6	12469	13	765184	20	8652429

These values are included for convenient indexing checks. The exact proof is provided by the preceding bijection, chamber reduction, and generating function, not by this table.

References

- [1] R. H. Hardin, *A195806: Number of triangular size-five arrays with entries $0, \dots, n$, equal sums on lines of equal length, and zero corners*, The On-Line Encyclopedia of Integer Sequences, introduced September 23, 2011. Conjectured recurrence and quasipolynomial contributed by M. Kauers and C. Koutschan, March 1, 2023. <https://oeis.org/A195806>. Retrieved September 20, 2026.
- [2] M. Kauers and C. Koutschan, *Some D-Finite and Some Possibly D-Finite Sequences in the OEIS*, Journal of Integer Sequences **26** (2023), Article 23.4.5, 42 pp. See Section 5.1, especially Conjecture 11 on p. 25. <https://cs.uwaterloo.ca/journals/JIS/VOL26/Koutschan/kout4.pdf>. Also arXiv:2303.02793, version 2, April 24, 2023, <https://arxiv.org/abs/2303.02793>. Journal and arXiv versions inspected September 20, 2026.